Associations Between Baseline Hard Drug Use and Two-Year Treatment Outcomes After HAART Initiation

Ezra Weible[1]

Introduction

The Multicenter AIDS Cohort Study (MACS) is an ongoing prospective study of HIV-1 infection among homosexual and bisexual men in four major US cities. This report analyzed a subset of 463 MACS participants with complete outcomes of interest (viral load (log10 HIV RNA copies/mL), CD4+ T cell count, and SF-36 aggregate physical and mental quality of life) at baseline (last untreated visit prior to HAART initiation) and two years after starting HAART. Baseline information on hard drug use, defined as injection drugs or illicit heroin/opiate use, was available along with demographic and behavioral covariates including age, BMI, smoking status, education, and race/ethnicity. Second year HAART adherence was also provided.

The primary question of interest was whether individuals who reported hard drug use at baseline experienced different clinical or quality of life outcomes two years after HAART initiation compared to those who did not. A secondary question was whether second year adherence could account for any observed differences. These translate into two statistical tasks: (1) Estimating the association between baseline hard drug use and each second year

[1]Biostatistics Masters Student, Colorado School of Public Health, CU Anschutz Medical Campus

outcome using linear regression models adjusted for baseline covariates and the corresponding baseline outcome in both frequentist and Bayesian frameworks, and (2) Re-estimating these associations with second year adherence included as an additional covariate and comparing the estimates to descriptively assess whether adherence explains the observed patterns.

Methods

All analyses were conducted in R v4.5.2 (2025-10-31 ucrt). The original dataset provided by the investigator included 715 individuals at baseline. Implausible BMI values (-1, 999, or >250) were set to missing. Categorical covariates were collapsed as follows to ensure adequate sample sizes: smoking status (non or former vs. current), education status (no college degree vs college degree and above), race/ethnicity (white, non-Hispanic vs other), and adherence (95% and above vs less than 95%). Viral load was transformed to the log10 scale to improve normality. Participants missing baseline covariates, baseline or second year outcomes, or second year adherence were removed, yielding a final sample of 463 individuals used for all models to ensure comparability across outcomes.

Primary Analysis: For each second year outcome, a linear regression was fit with baseline hard drug use as the primary predictor, adjusting for age, BMI, smoking status, education status, and race/ethnicity. Each model additionally adjusted for the baseline value of the corresponding outcome to account for pre-treatment differences. Frequentist models reported point estimates, 95% confidence intervals (CIs), and unadjusted p-values. All four outcomes were prespecified as equally important, so no adjustments were applied.

Bayesian models were implemented in STAN with vaguely informative priors (N(0,1000) for regression coefficients, half-normal(0,1000) for residual SD), reflecting limited prior knowledge. The intercept was not centered due to the large variance. Sampling used the Hamiltonian Monte Carlo (HMC) algorithm with the no-U-turn sampler (NUTS). Models were fit using four chains with 5000 iterations following 2000 burn-in with a fixed seed for reproducibility. Convergence and mixing were assessed with R-hat, bulk and tail effective sample sizes, trace plots, autocorrelation plots, and posterior density plots. All diagnostics indicated good convergence and adequate mixing across all models. Posterior means, 95% highest posterior density intervals (HPDIs), and posterior probabilities were reported. Posterior probabilities were based on clinically meaningful thresholds for each outcome (two-sided): 0.5 log10 (viral load), 50 cells (CD4+ T cell count), and 2 points (quality of life scores).

Adherence-Adjusted Models: To evaluate whether second year adherence explained any observed associations, each model was refit to include adherence. Estimates were then compared descriptively to the primary models.

Because only one hard drug user had adherence <95%, a sensitivity analysis used an alternative adherence definition (<100% vs. 100%). This increased the number of suboptimally adherent hard drug users from 1 to 18, improving model stability. Bayesian and frequentist estimates under this alternative definition were nearly identical to those observed using the investigator-specified cutoff, so the original definition was retained for clinical relevance.

Results

Table 1 summaries baseline characteristics and outcomes by hard drug use, and *Table 2* summarizes year 2 outcomes and adherence. *Figure 1* visualizes the distributions of year 2 outcomes by hard drug status.

Primary Analysis: Frequentist and Bayesian estimates were highly consistent (*Table 3*), Bayesian results are emphasized in this section. For viral load, the posterior mean difference associated with baseline hard drug use was -0.057 (95% HPDI: -0.454, 0.343), with a posterior probability of 0.017 for a clinically meaningful effect. This indicates substantial uncertainty and little evidence of an association.

For CD4+ T cell count, the posterior mean difference was -163.66 cells (95% HPDI: -225.11, -99.97). The posterior probability of a clinically meaningful effect was 1.00 and the entire 95% HPDI exceeded the clinically meaningful threshold of 50 cells, indicating a reduction in CD4+ T cell count among individuals who reported baseline hard drug use.

For physical quality of life, the posterior mean was -3.34 (95% HPDI: -6.03, -0.59) with a posterior probability of 0.83, suggesting there is moderate evidence of reduced physical quality of life among baseline hard drug users.

For mental quality of life, the posterior mean (-0.28, 95% HPDI: -3.55, 3.36) and posterior probability (0.259) provide little evidence of association.

Adherence-Adjusted Models: Adjustments for second year adherence produced minimal changes in the effect estimates (*Table 4*). For viral load, the adherence-adjusted posterior mean had a change of 77.6% from -0.057 to -0.013 (95% HPDI: -0.417, 0.377) and the posterior probability remained low (0.014). For CD4+ T cell count, the adherence-adjusted posterior mean differed by only 3.78% and the posterior probability remained 1.00. For physical quality of life, the posterior mean had a -6.92% change from the unadjusted model, and the posterior probability modestly increased to 0.870 from 0.833. For mental quality of life, the adherence-adjusted posterior mean was attenuated by 94.0% relative to the unadjusted model from -0.28 to -0.54 (95% HPDI: -3.99, 2.91) and the posterior probability remained low (0.275). Across all outcomes, adjusting for adherence did not alter conclusions.

Figure 2 illustrates the close agreement between frequentist and Bayesian estimates with and without adherence included in the model.

Discussion

This paper evaluated whether baseline hard drug use was associated with clinical and quality of life outcomes two years after HAART initiation. Hard drug use was strongly associated with lower CD4+ T cell counts and moderately associated with lower physical quality of life scores at year 2, after adjusting for baseline covariates. Associations with viral load and mental quality of life were small and highly uncertain. Bayesian and frequentist results were closely aligned, and posterior probabilities only indicated strong support for the CD4+ association.

Including second year adherence into the models produced minimal changes in effect estimates or uncertainty, and a sensitivity analysis using an alternative adherence definition yielded nearly identical results. These findings suggest that the observed associations between baseline hard drug use and worse CD4+ T cell counts and physical quality of life are not explained by the available adherence measure.

Several limitations should be considered. Hard drug use and adherence were self-reported, which may introduce measurement error. Further, they were only included from a single time point, limiting the ability to capture variation that may influence outcomes. Collapsing categories due to small sample size may have reduced power. Complete-case analysis may introduce bias if missingness is related to unmeasured characteristics associated with both exposure and outcomes. Differential dropout or missingness related to hard drug use could bias estimates. Important potential confounders such as socioeconomic status, mental health conditions, or co-infections were not included. The results reflect the structure of the MACS cohort and may not generalize to other populations. These limitations should be taken into account when interpreting these estimated associations.

Appendix

Code Directory: https://github.com/eweible/BIOS6624.git

File name: P1/Code/P1_Report.Rmd

Data Directory: C:/Users/weiblee/OneDrive/Documents/Anschutz/Spring 2026/6624_Advanced/Weible_Advanced/P1/DataRaw

Table 1: Baseline Characteristics by Baseline Hard Drug Use

	No (N=427)	**Yes (N=36)**	**Overall (N=463)**
Viral Load (log10 HIV copies/mL blood)			
Mean (SD)	4.53 (0.930)	4.58 (0.871)	4.53 (0.925)
Median [Min, Max]	4.52 [0.237, 8.28]	4.47 [2.87, 6.40]	4.51 [0.237, 8.28]
CD4+ T Cell Count			
Mean (SD)	374 (202)	360 (201)	373 (202)
Median [Min, Max]	360 [10.9, 1220]	397 [10.9, 650]	360 [10.9, 1220]
Physical Quality of Life (SF-36 Aggregate)			
Mean (SD)	51.4 (9.00)	49.2 (7.05)	51.2 (8.88)
Median [Min, Max]	53.7 [19.2, 69.0]	47.4 [31.4, 62.9]	53.5 [19.2, 69.0]
Mental Quality of Life (SF-36 Aggregate)			
Mean (SD)	45.2 (13.8)	42.0 (11.6)	44.9 (13.6)
Median [Min, Max]	49.5 [7.23, 66.0]	44.4 [22.5, 59.6]	49.1 [7.23, 66.0]
Age (years)			
Mean (SD)	43.1 (8.68)	43.9 (9.52)	43.2 (8.74)
Median [Min, Max]	43.0 [20.0, 73.0]	45.5 [29.0, 61.0]	43.0 [20.0, 73.0]
BMI (kg/m^2)			
Mean (SD)	25.3 (4.39)	23.6 (3.45)	25.2 (4.34)
Median [Min, Max]	24.8 [16.5, 45.3]	23.3 [18.0, 31.2]	24.6 [16.5, 45.3]
Smoking Status			
Non or Former Smoker	278 (65.1%)	9 (25.0%)	287 (62.0%)
Current Smoker	149 (34.9%)	27 (75.0%)	176 (38.0%)
Education Status			
No College Degree	236 (55.3%)	26 (72.2%)	262 (56.6%)
College Degree and above	191 (44.7%)	10 (27.8%)	201 (43.4%)
Race/Ethnicity			
White, non-Hispanic	277 (64.9%)	19 (52.8%)	296 (63.9%)
Other	150 (35.1%)	17 (47.2%)	167 (36.1%)

Table 2: Year 2 Characteristics by Baseline Hard Drug Use

	No (N=427)	**Yes (N=36)**	**Overall (N=463)**
Viral Load (log10 HIV copies/mL blood)			
Mean (SD)	1.79 (1.21)	1.85 (1.45)	1.79 (1.23)
Median [Min, Max]	1.49 [-0.608, 5.85]	1.64 [-0.131, 5.63]	1.49 [-0.608, 5.85]
CD4+ T Cell Count			
Mean (SD)	556 (264)	371 (249)	542 (267)
Median [Min, Max]	536 [39.5, 1730]	357 [60.0, 971]	516 [39.5, 1730]
Physical Quality of Life (SF-36 Aggregate)			
Mean (SD)	50.0 (9.93)	44.4 (12.0)	49.6 (10.2)
Median [Min, Max]	53.3 [14.8, 69.1]	44.8 [18.2, 63.9]	53.2 [14.8, 69.1]
Mental Quality of Life (SF-36 Aggregate)			
Mean (SD)	47.6 (11.8)	45.9 (14.1)	47.4 (12.0)
Median [Min, Max]	51.2 [10.5, 66.7]	49.2 [21.3, 65.3]	51.2 [10.5, 66.7]
Adherence to HAART since Last Visit			
95% and Above	382 (89.5%)	35 (97.2%)	417 (90.1%)
<95%	45 (10.5%)	1 (2.8%)	46 (9.9%)

Table 3: Comparison of Frequentist and Bayesian Estimates for Associations Between Hard Drug Use and Clinical Outcomes						
	Frequentist			Bayesian		
Outcome	Point Estimate	95% CI	Unadjusted P-Value	Posterior Mean	95% HPDI	Posterior Probability
Viral Load (log10 HIV copies/mL blood)	−0.059	(-0.456, 0.337)	0.770	−0.057	(-0.454, 0.343)	0.017
CD4+ T Cell Count	−163.880	(-226.180, -101.580)	0.000	−163.660	(-225.105, -99.968)	1.000
Physical Quality of Life (SF-36 Aggregate)	−3.336	(-6.055, -0.617)	0.017	−3.341	(-6.028, -0.593)	0.833
Mental Quality of Life (SF-36 Aggregate)	−0.289	(-3.743, 3.165)	0.870	−0.278	(-3.551, 3.356)	0.259

Comparison of Frequentist and Bayesian regression estimates for the association between baseline hard drug use and year-2 outcomes. All models adjust for baseline outcome values, age, BMI, smoking status, education, and race. Bayesian models use weakly informative Normal(0, 1000) priors for regression coefficients and half-Normal(0, 1000) priors for residual SD.

Table 4: Impact of Adherence Adjustment on Frequentist and Bayesian Estimates for Associations Between Hard Drug Use and Clinical Outcomes

	Frequentist				Bayesian			
Outcome	Point Estimate	Change in PE (%)	95% CI	Unadjusted P-Value	Posterior Mean	Change in PM (%)	95% HPDI	Posterior Probability
Viral Load (log10 HIV copies/mL blood)	−0.013	78.603	(-0.409, 0.384)	0.950	−0.013	77.617	(-0.417, 0.377)	0.014
CD4+ T Cell Count	−169.683	−3.541	(-232.049, -107.317)	0.000	−169.849	−3.781	(-231.897, -107.217)	1.000
Physical Quality of Life (SF-36 Aggregate)	−3.567	−6.924	(-6.291, -0.843)	0.011	−3.573	−6.923	(-6.273, -0.812)	0.870
Mental Quality of Life (SF-36 Aggregate)	−0.531	−83.958	(-3.995, 2.933)	0.764	−0.540	−94.006	(-3.985, 2.909)	0.275

Frequentist and Bayesian regression estimates for the association between baseline hard drug use and year-2 outcomes after adjusting for adherence. All models also adjust for baseline outcome values, age, BMI, smoking status, education, and race. Percent change reflects the relative difference between adherence-adjusted and non-adjusted models, calculated for both the frequentist point estimate and the Bayesian posterior mean. Bayesian models use weakly informative Normal(0, 1000) priors for regression coefficients and half-Normal(0, 1000) priors for residual SD.

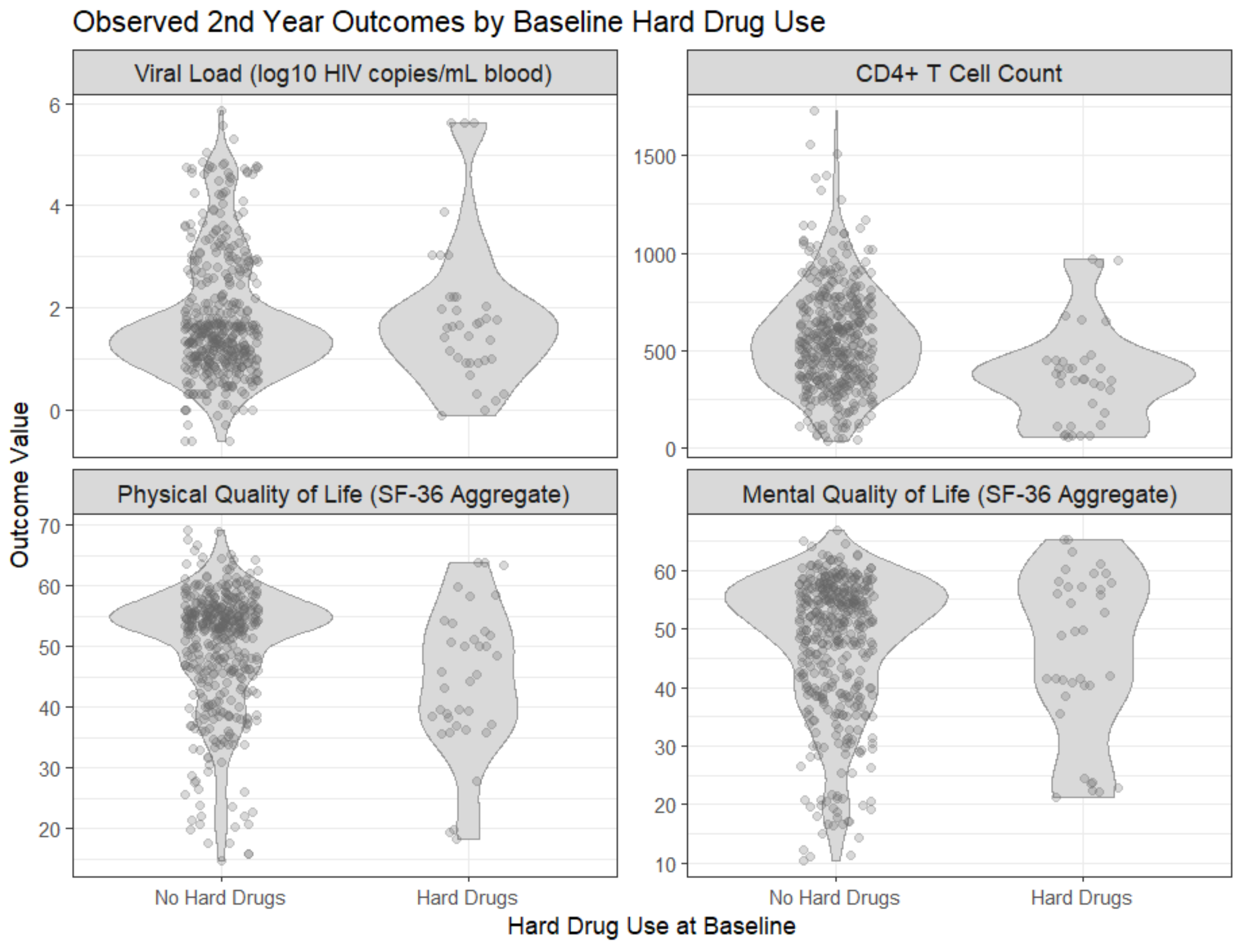

Figure 1: Year 2 observed outcomes by hard drug use at baseline.

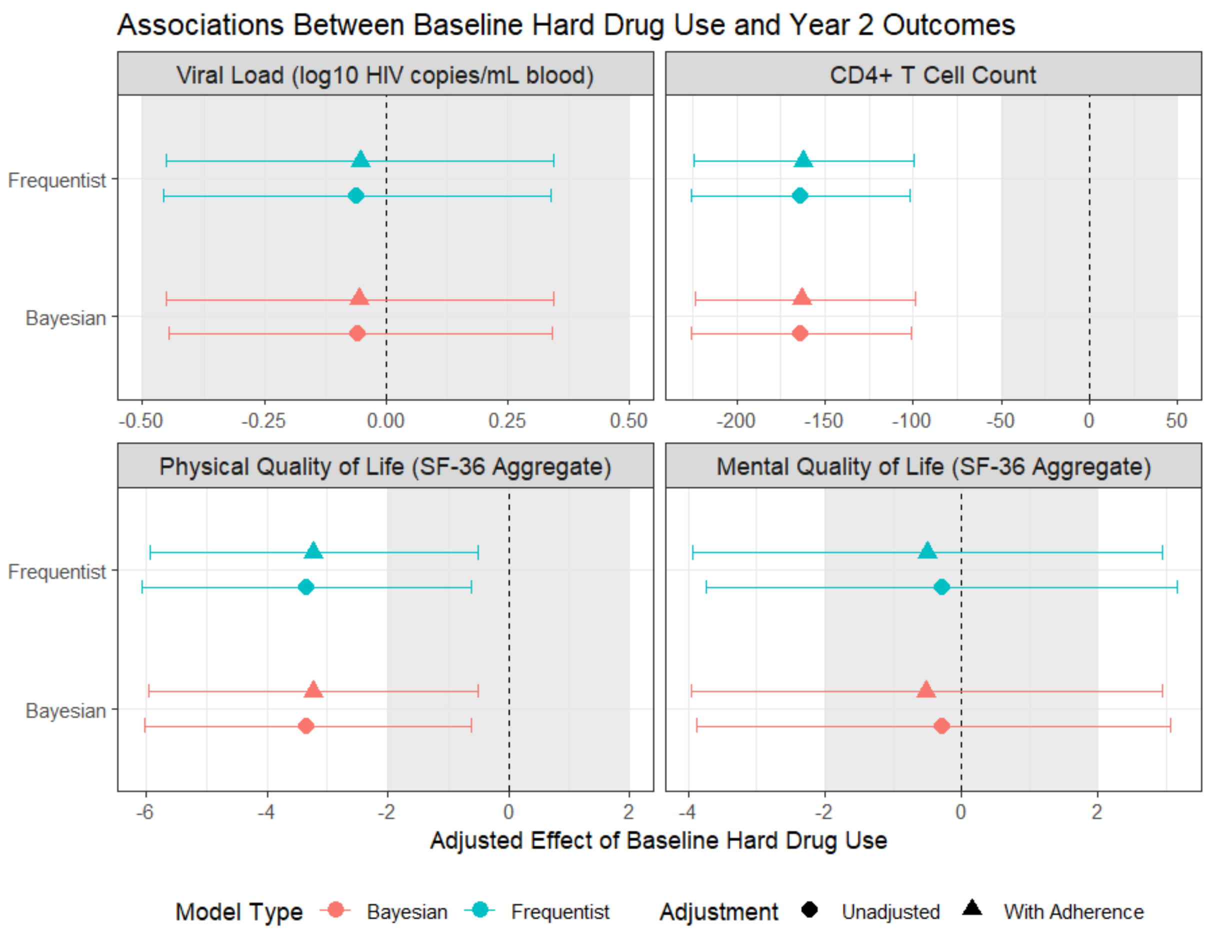

Figure 2: Adjusted associations between baseline hard drug use and 2nd year clinical outcomes estimated using Frequentist and Bayesian regression models, shown separately for models with and without adherence. All models adjust for baseline outcome values, age, BMI, smoking status, education, and race. Bayesian models use weakly informative Normal(0, 1000) priors for regression coefficients and half-Normal(0, 1000) priors for residual standard deviations. Points represent the estimated effect of baseline hard drug use, with error bars showing 95% confidence intervals (Frequentist) or 95% highest posterior density intervals (Bayesian). The grey shaded region in each panel denotes the range corresponding to the two-sided clinically meaningful difference for that outcome (0.5 log10 for viral load, 50 cells for CD4 count, and 2 points for quality of life scales), providing a visual benchmark for interpreting the magnitude of estimated effects.